



Thermodynamics II Report task (2025,5,16)

1. Answer the following questions.

- (1) Explain the second law of thermodynamics.
- (2) Describe both “free energy” and “bound energy”.
- (3) Give the defining equation of Gibbs free energy. Also, show the relation to the Helmholtz free energy.
- (4) Find the $\Delta_f G^\circ$ and $\Delta_f H^\circ - T\Delta S$ in the reaction ($C + O_2 \rightarrow CO_2$) in which carbon dioxide is produced at 298.15K from C and O_2 of the standard substance and confirm that they are equal. Please refer the table of $\Delta_f G^\circ$, $\Delta_f H^\circ$ and ΔS in the handout provided in the day 3.

2. Consider air at 300 K and $0.86 \text{ m}^3/\text{kg}$. The state of air changes to 302 K and $0.87 \text{ m}^3/\text{kg}$ as a result of some disturbance.

Using

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy,$$

estimate the change on the pressure of air $p(T, v)$.

Here, air is assumed as ideal gas, and $0.287 \text{ [kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}]$ can be used as the gas constant.

3. Find the change in Gibbs free energy ΔG , when 1 kg of ideal gas expands to 10 times its volume at constant temperature 25°C. Here, the gas constant R is 8.31 J/Kmol .

4. Answer the following questions.

- (1) Find the thermodynamic identity for the Helmholtz free energy and the Gibbs free energy.
- (2) From the thermodynamic identity obtained in (1), derive two expressions for Maxwell's thermodynamic relations.
- (3) If the system is an ideal gas, show that H is a function of T only. It can be derived by starting the calculation from $dH = TdS + Vdp$.

[Hint] From $dH = TdS + Vdp$, please derive

$$\left(\frac{\partial H}{\partial p}\right)_T = T\left(\frac{\partial S}{\partial p}\right)_T + V = -T\left(\frac{\partial V}{\partial T}\right)_p + V$$

first.